



The effect of time-dependent level kinetics in a 1-D MHD simulation of an argon gas z-pinch implosion*

N. D. Quart, A. Dasgupta, A. L. Velikovich, J. L. Giuliani[†], and V. Tangri

Plasma Physics Division, U.S. Naval Research Laboratory, Washington, DC 20375 USA

[†]NRL Voluntary Emeritus Program

M.-A. Schaeuble, J. Schwarz, D. Ampleford, R. A. Vesey, C. Jennings, and B. Jones

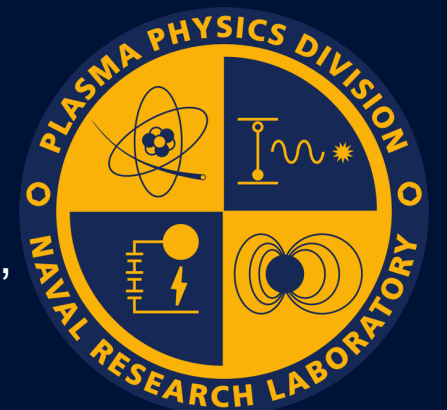
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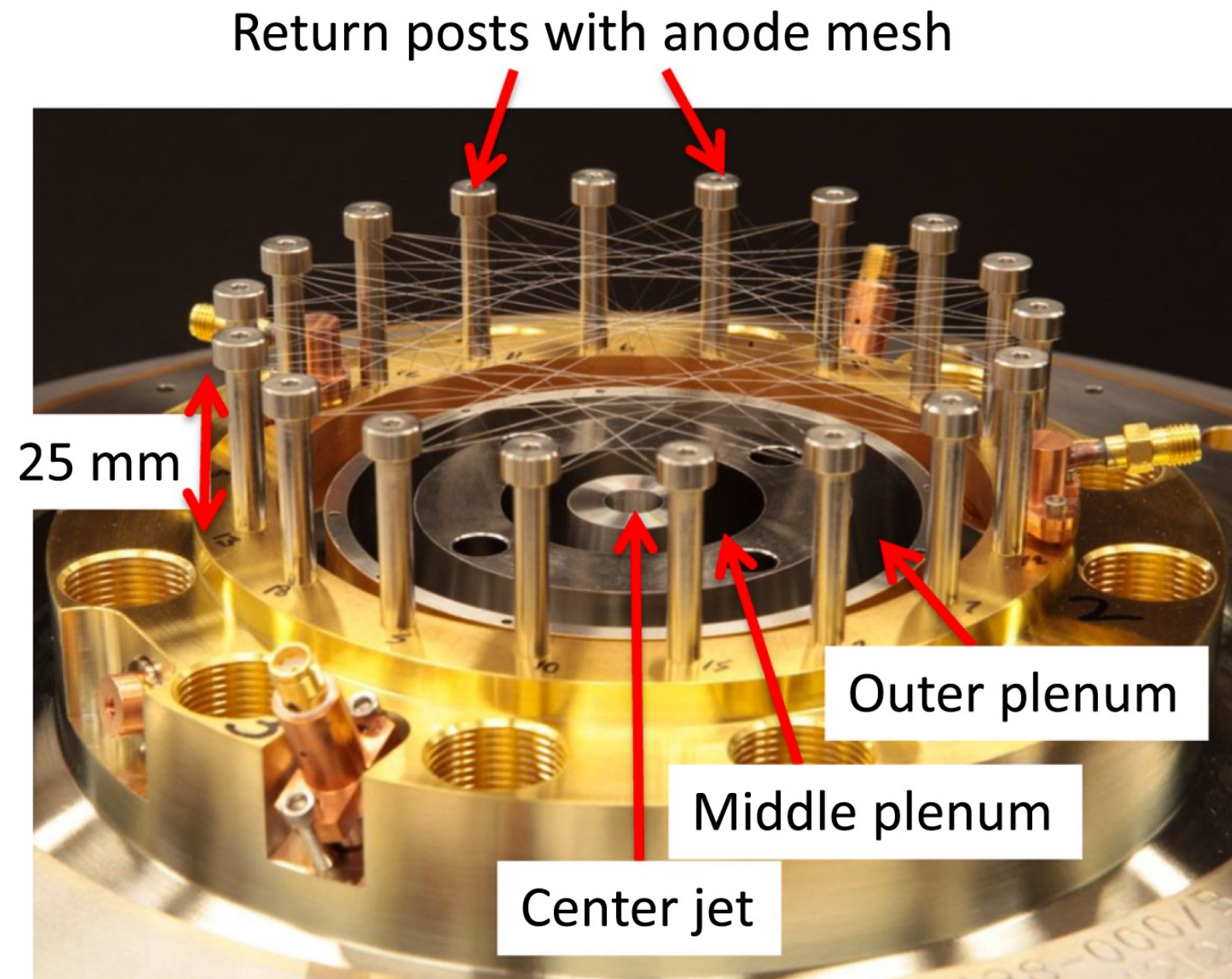
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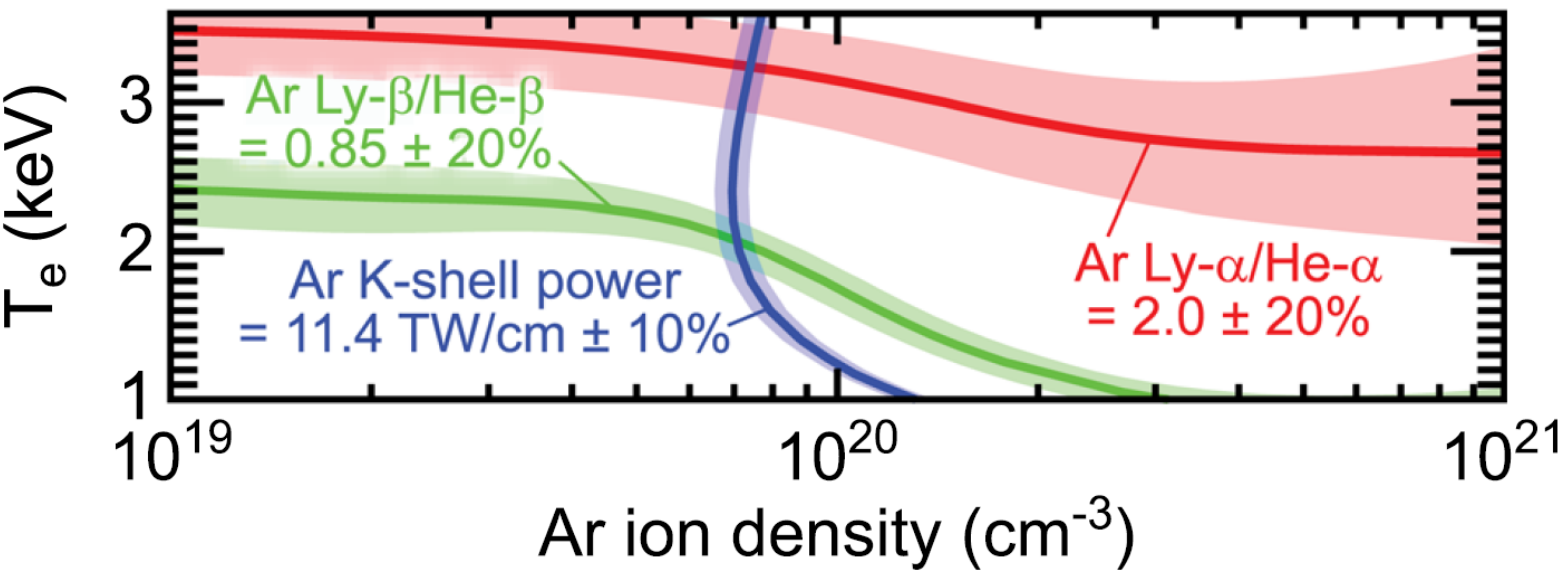
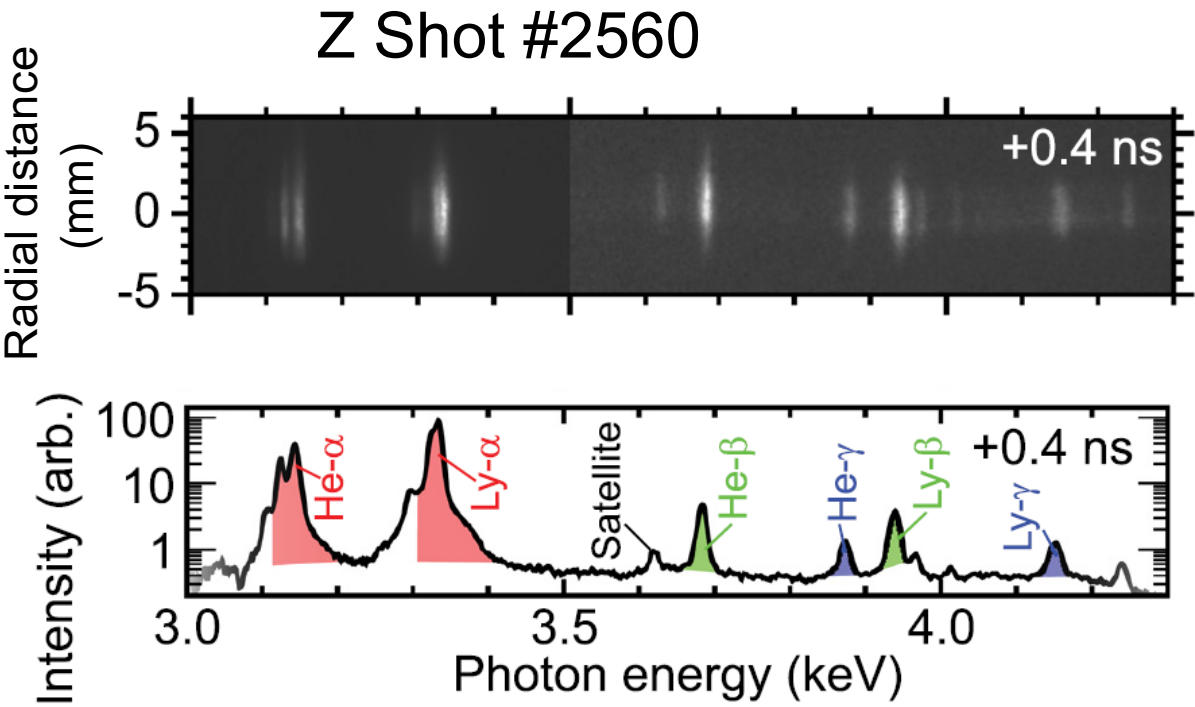
- Argon gas puff experiments on the Z machine have produced $330 \text{ kJ} \pm 9\%$ above 3 keV photon energy.
- Spectroscopy was used to infer the plasma conditions near peak power
- Mach2-TCRE simulations reproduced the radiative powers and yields from Z Shot 2560
 - Ly- α /He- α +IC line ratio had good agreement after peak power, but was higher prior to peak power.
 - This was attributed to 3-D effects and the assumption of steady-state populations in TCRE.
- The 1-D NRL DZAPP code is used to assess the effects of time-dependent rate equations for the ground state level populations compared to steady-state for the Ly- α /He- α +IC line ratio
 - There was a decrease in the line ratio in the case of the time-dependent atomic level kinetics case
 - However, the line ratio was still larger compared with experiment.
 - Time-dependent level kinetics and multi-dimensional effects may both be important.
- Future work will assess the radiative cooling caused by Doppler shifts, due to large radial velocity gradients, on the line ratio in DZAPP.

Argon gas puff experiments on the Z machine have produced $330 \text{ kJ} \pm 9\%$ above 3 keV photon energies



- An 8 cm diameter double-shell gas nozzle was used
- The mass of the argon gas was 1 mg/cm.
- The inner- to outer-shell mass ratio was 1.6:1.
- The length was 2.5 cm.

The Ar Lyβ line intensity is reduced by inner-shell photoionization in the cooler region

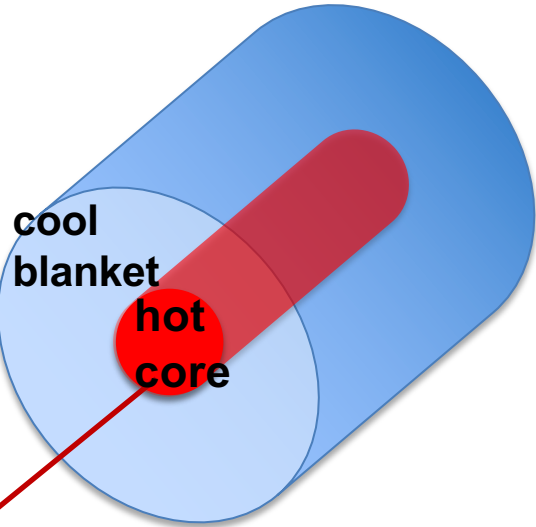


	experiment	Two region model
Ar Lyα/Heα	2.00±20%	1.54
Ar Lyβ/Heβ	0.85±20%	0.91
Ar Lyy/Hey	1.20±20%	1.32
K-shell power (TW/cm)	11.4±10%	11.5
Total power (TW/cm)	16±20%	16.0
Mass (mg/cm)	1±10%	1.01

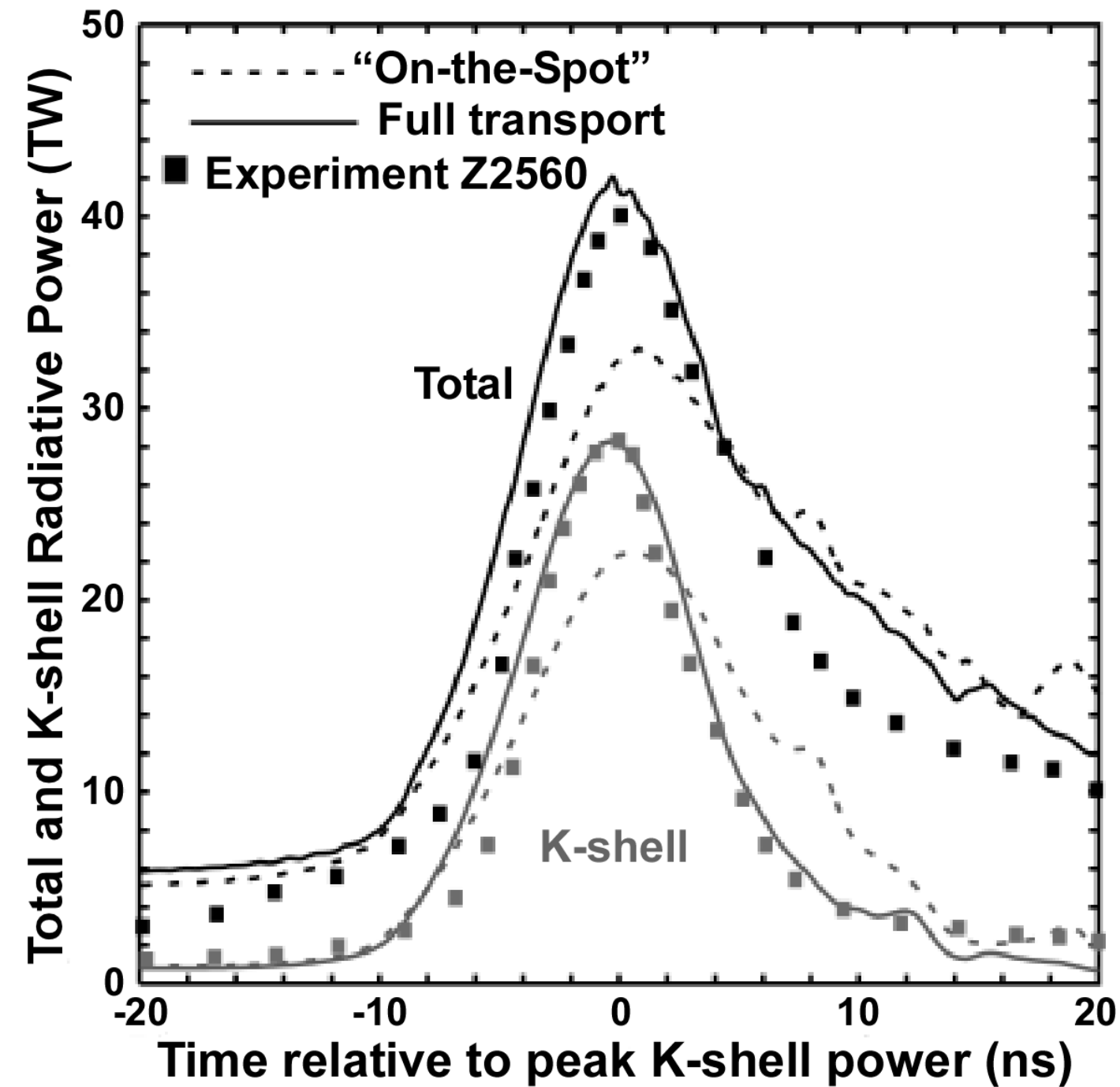
χ^2 is 1.16 with a confidence level of 98%

L-shell region
 $n_i = 2.5 \times 10^{19} \text{ cm}^{-3}$
 $T_e = 110 \text{ eV}$
 $\Delta r = 2.6 \text{ mm}$
73% total mass

K-shell region
 $n_i = 6.7 \times 10^{19} \text{ cm}^{-3}$
 $T_e = 2.45 \text{ keV}$
Ø2.8mm
27% total mass

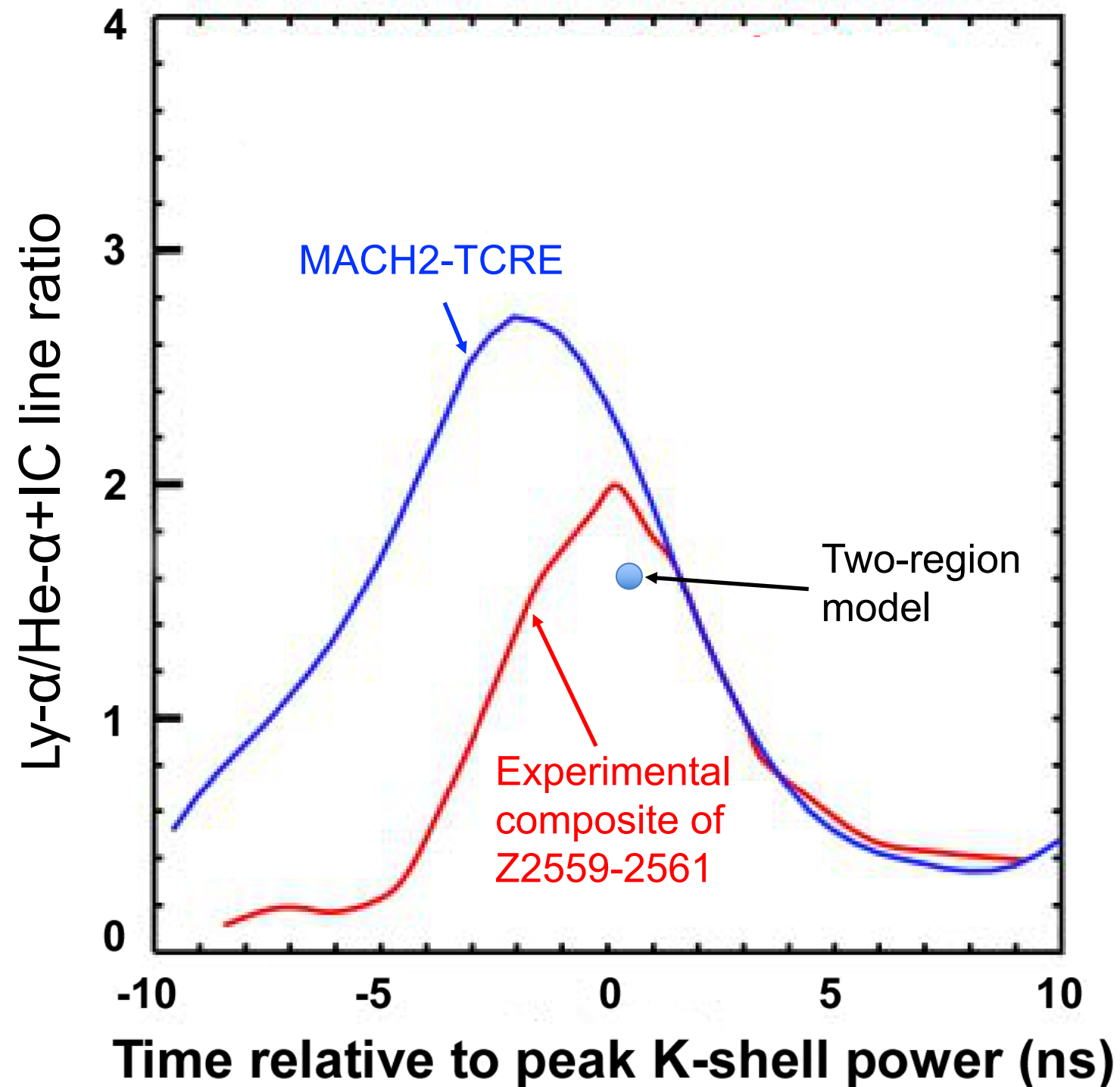


Simulations using Mach2-TCRE were able to reproduce the radiative powers and yield from shot Z2560



J.W. Thornhill *et al.* IEEE TPS 43, 2480 (2015)

The simulation's Ly- α /He- α +IC line ratio had good agreement after peak power, but was higher prior to peak power.



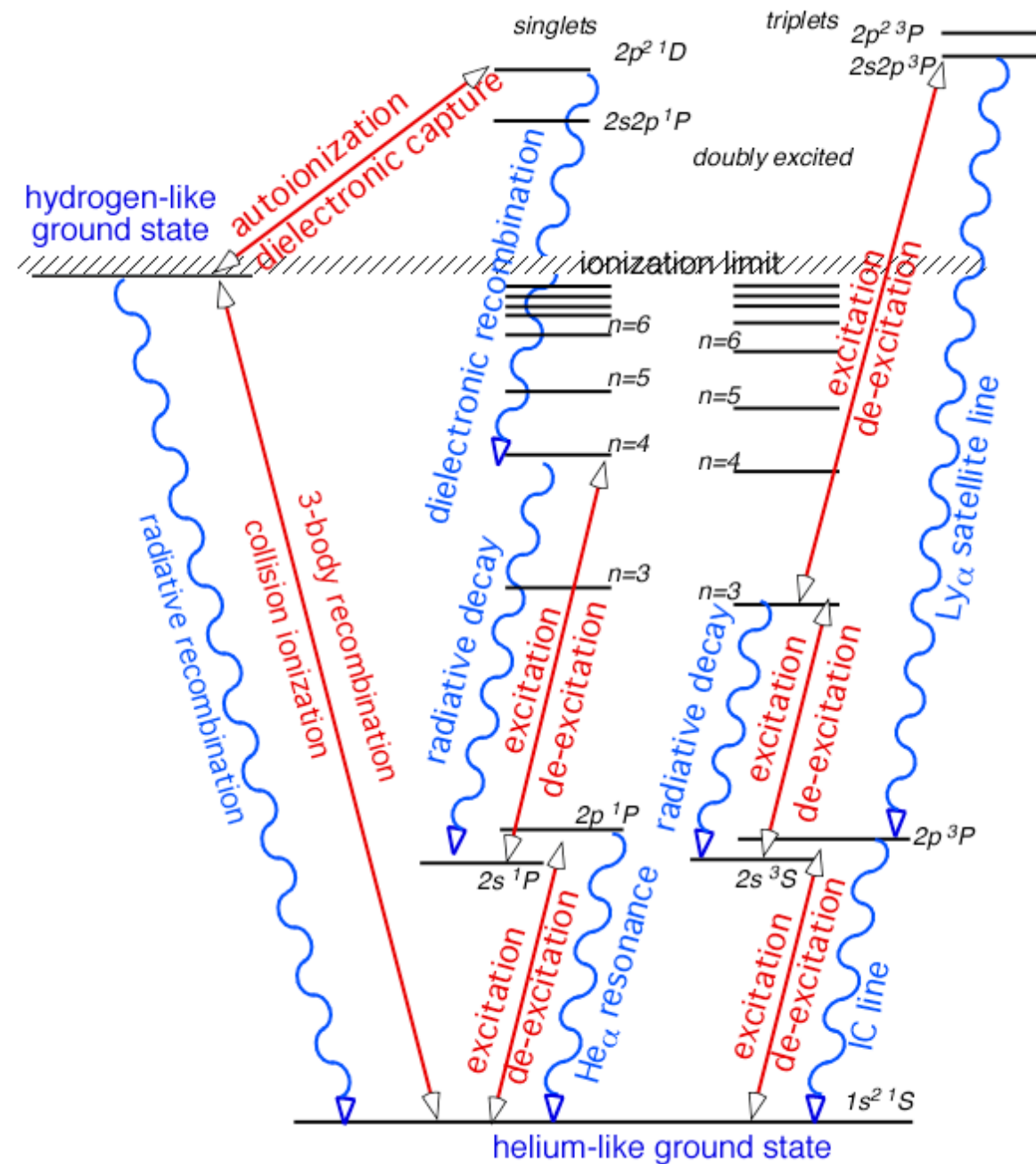
Label	Upper	Lower	Energy
IC	1s2p 3P_1	1s 2 1S_0	3.124 keV
He- α	1s2p 1P_1	1s 2 1S_0	3.140 keV
Ly- α_2	2p $^2P_{1/2}$	1s $^2S_{1/2}$	3.318 keV
Ly- α_1	2p $^2P_{3/2}$	1s $^2S_{1/2}$	3.323 keV

Thornhill *et al.* attributed this discrepancy prior to peak power to:

1. 3-D effects
2. Implicit assumption of steady-state populations in the TCRE table.

This presentation will explore the effect of time-dependent level kinetics on the line ratio.

The NRL DZAPP code was used to assess the effect of time-dependent and steady-state atomic level kinetics



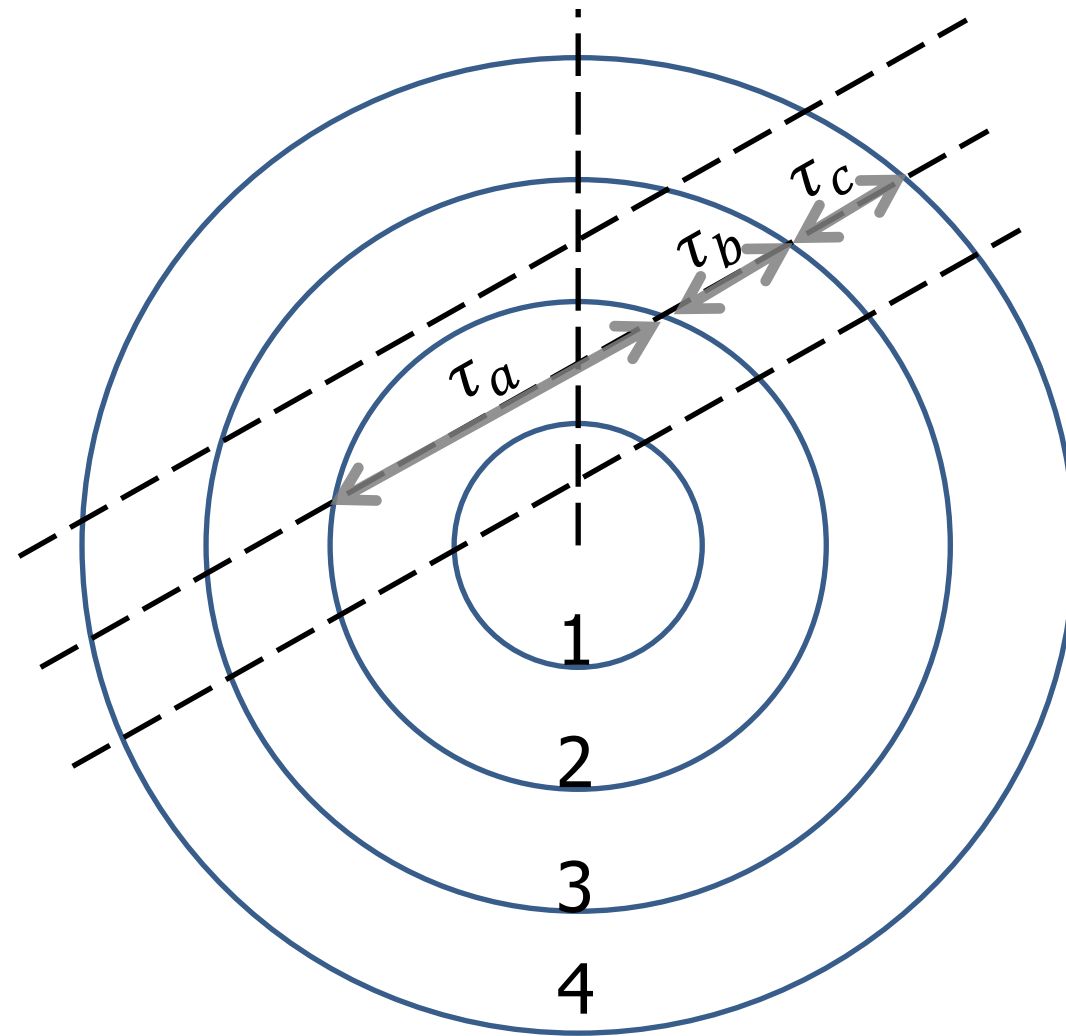
The populations depend on the rates of these atomic processes and can be determined by:

$$\frac{df_{ik}}{dt} = \sum_j (W_{jik}f_{jk} - W_{ijk}f_{ik}) + \text{photo-pumping by non-local radiation field}$$

The time derivative is set to zero for steady-state populations

All physical processes in our collisional-radiative modeling are driven by non-LTE atomic kinetics

Multi-zone radiation transport via the coupling constant formalism provides good overall energetics at a reasonable cost in computer time



Multi-zone radiation transport is necessary because photons are more likely to escape near the edge

$$C_{24} = \frac{1}{\tau_a} \int_0^{\tau_a} [P_e(\tau_b + \tau) - P_e(\tau_b + \tau_c + \tau)] d\tau$$

Reciprocity relation:

$$C_{42} = C_{24} \frac{N_2}{N_4}$$

Ratio of number
of absorbers

The photo-excitation rate for transition $i'-i$ in spatial zone k :

$$W_{ii'k} = 4\pi \sum_{k' \neq k} C_{k'k}^{ii'} j_{k'}^{ii'} \frac{V_{k'}}{N_k f_{ik} E_{ii'} V_k}$$

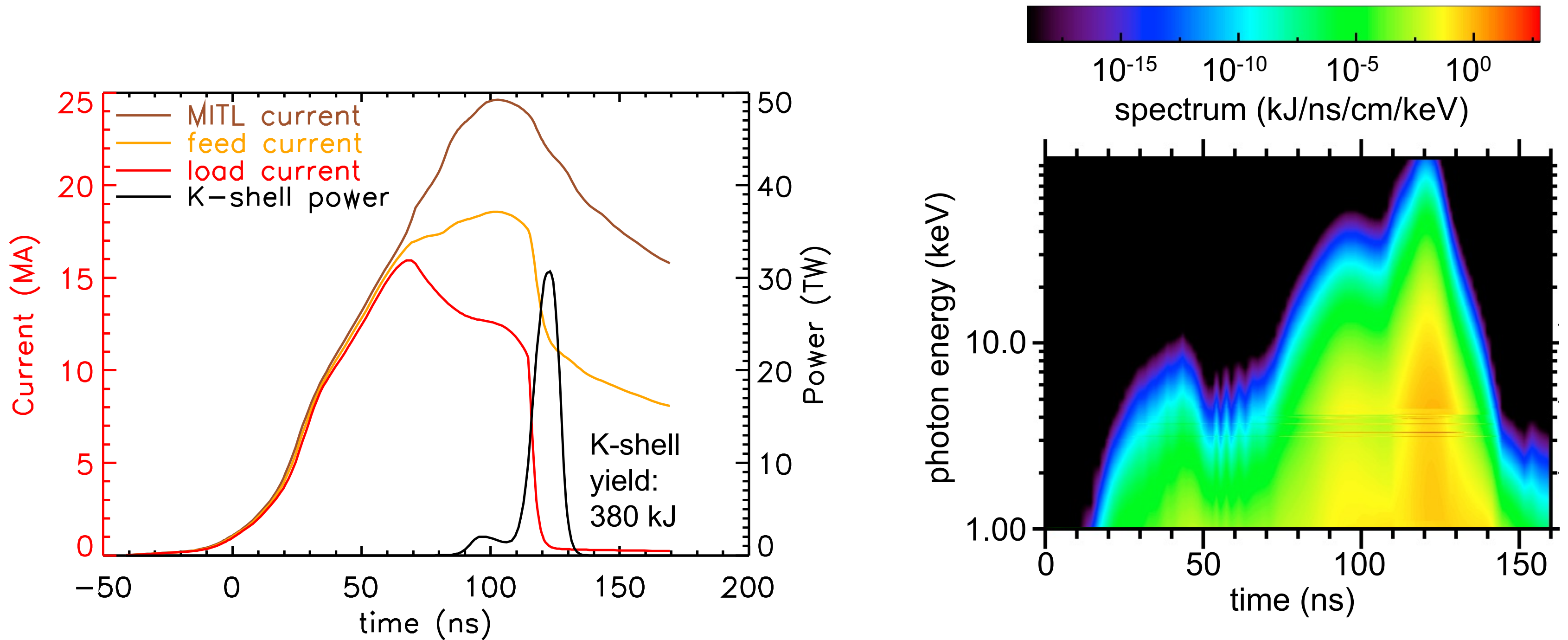
J.P. Apruzese *et al.* JQSRT 23, pp. 479-487 (1980)

J.P. Apruzese. JQSRT 25, pp. 419-425 (1981)

J.P. Apruzese. JQSRT 34, pp. 447-452 (1985)

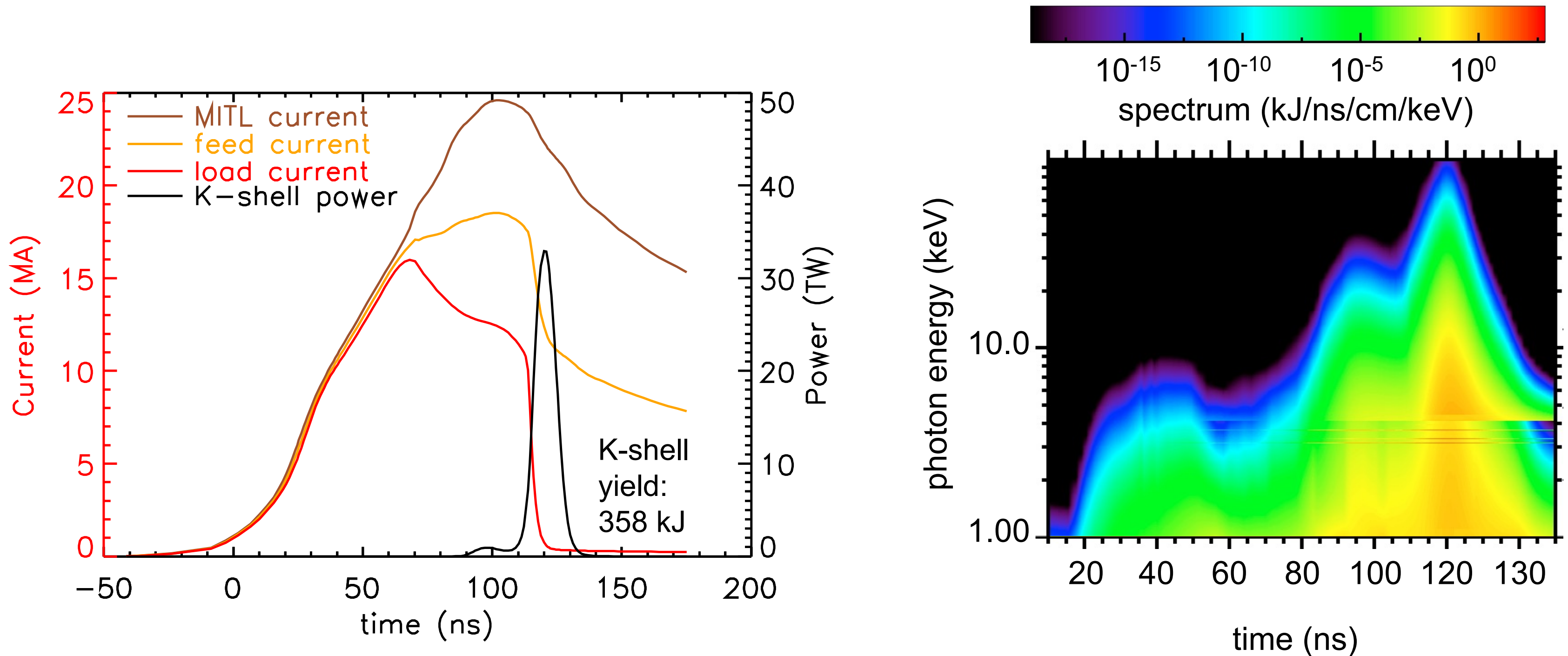
R.W. Clark *et al.* JQSRT **53**, pp. 307-320 (1995)

Current waveforms, x-ray powers, and spectra assuming collisional-radiative steady-state level kinetics



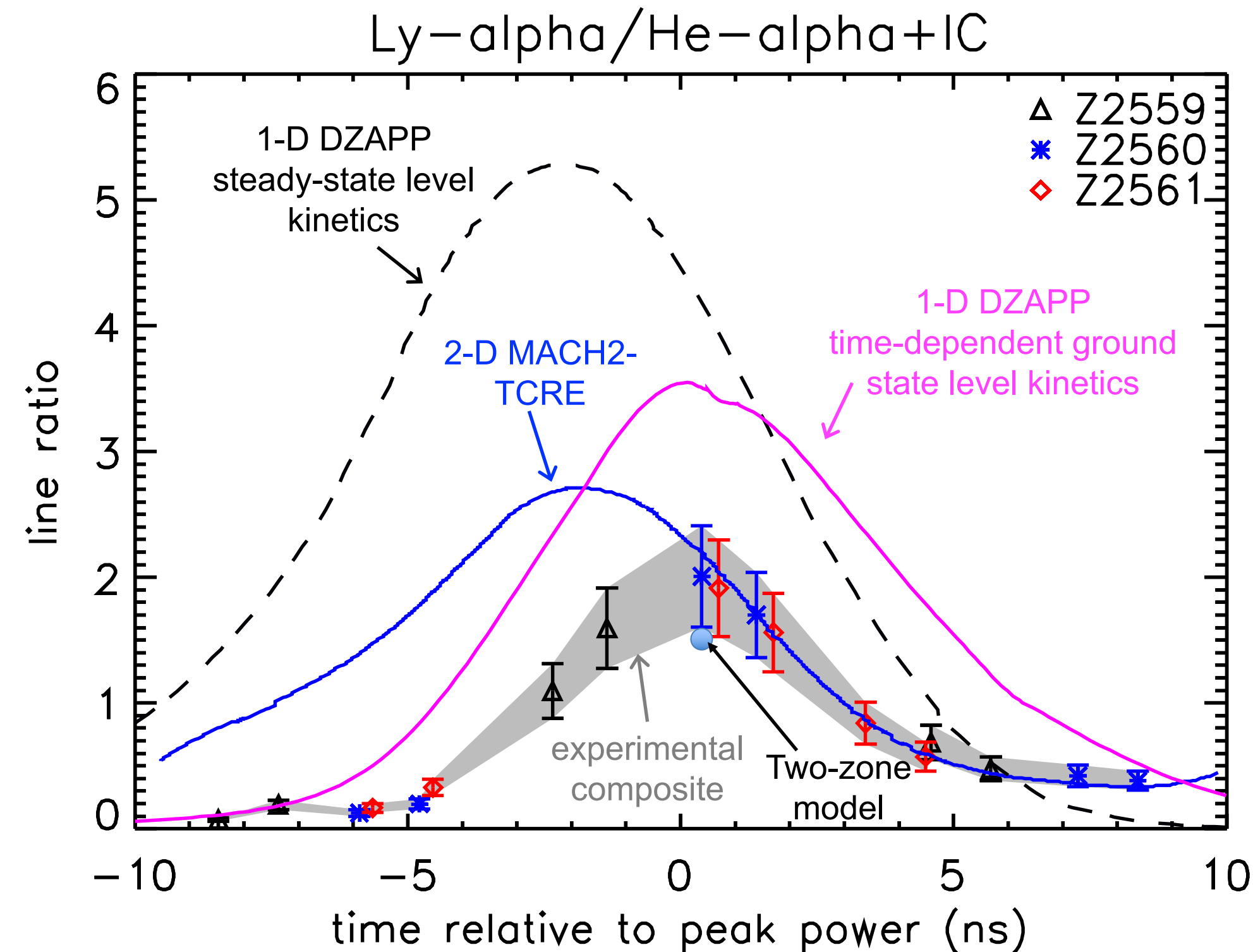
Z shot #2560 had a K-shell yield of $363 \text{ kJ} \pm 8\%$ and a peak K-shell power of 29 TW

Current waveforms, x-ray powers, and spectra assuming time-dependent rate equations for ground state level populations



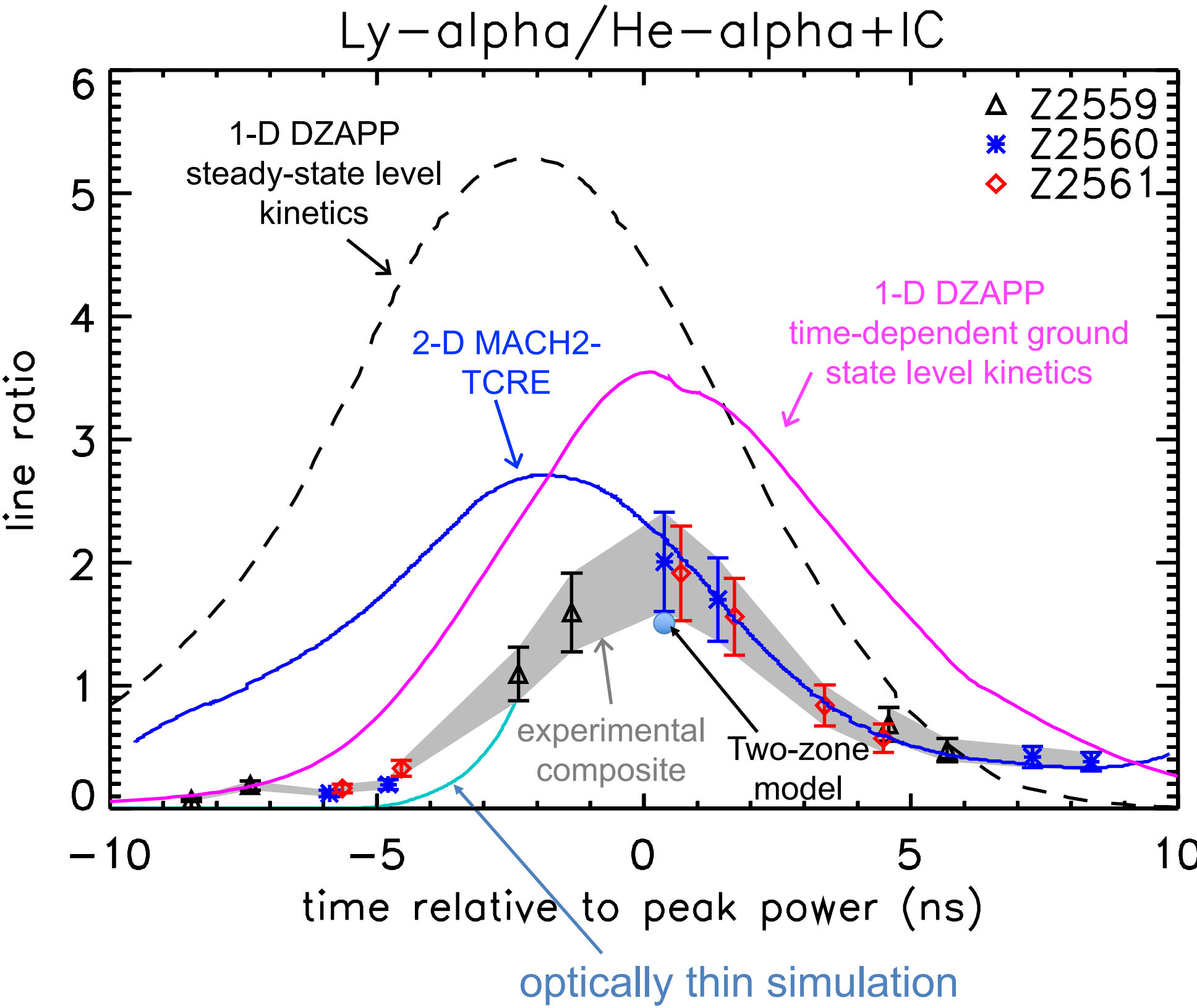
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There is a downward trend in the peak line ratio and FWHM when using the time-dependent ground state level kinetics

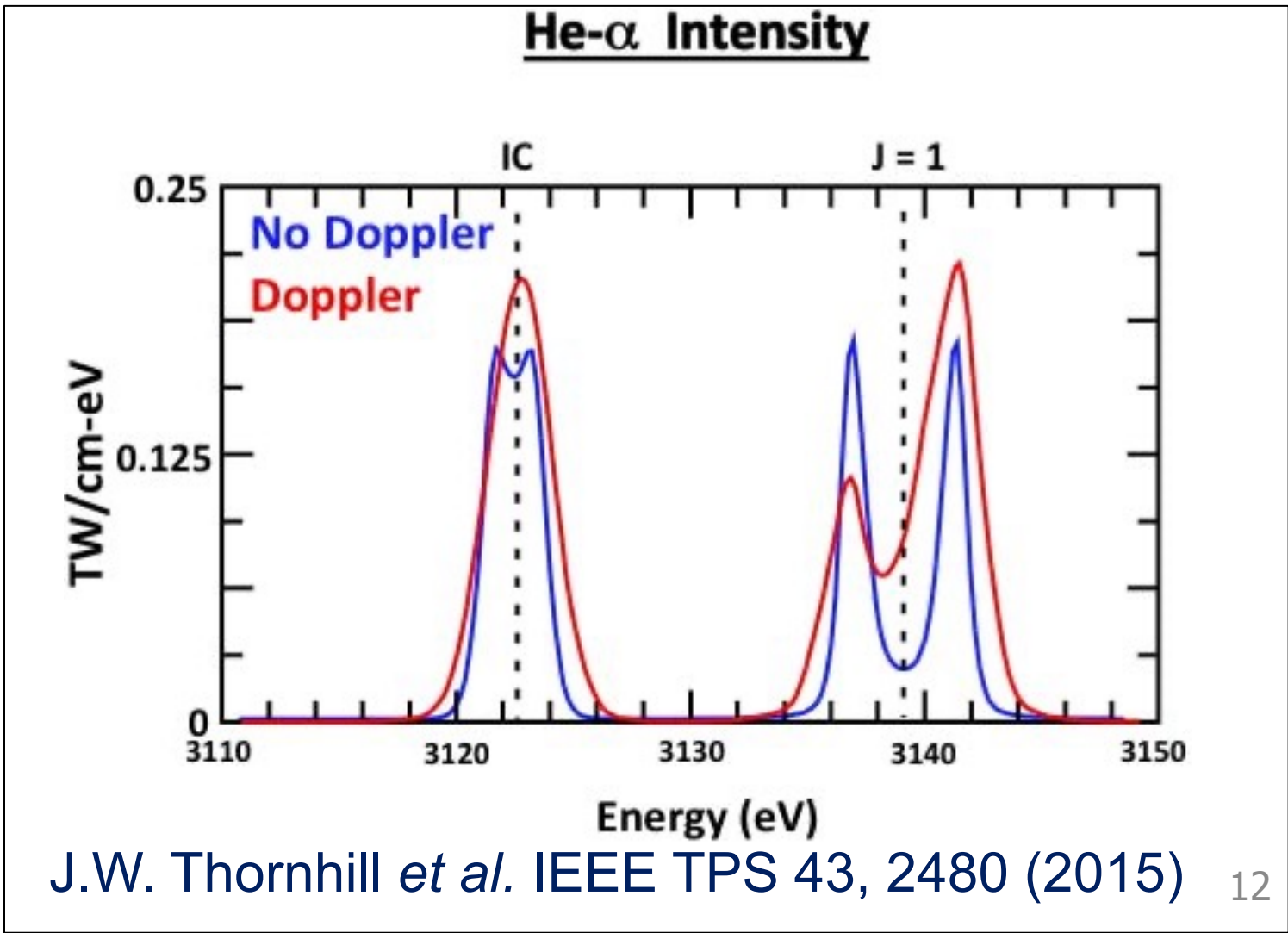


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Including Doppler shifts were shown to reduce the absorption of the He- α and IC lines.



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He- α	$1s2p\ ^1P_1$	$1s^2\ ^1S_0$	3.140 keV
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